Evaluating performance of fine-tuned benchmark CNNs and geometry encoded MLP for sparse crowd counting applications

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I. Abstract

Two large issues regarding current convolutional neural networks (CNNs) are both domain shift and data privacy. For this, we will compare state-of-the-art models with an approach that compresses the overall image difference into compressed data that is the input into a simpler model. We also highlight how including geometric encoding of the specified environment is suitable for improving the accuracy of a system at the cost of generalization.

II. Introduction

A. Background

Crowd counting has evolved from early detection-based methods [1] to regression-based methods [2], and subsequently to modern deep learning approaches. CNN-based methods further advance the task by learning spatial representations directly from images. Among them, density-map-based models such as CSRNet [3] estimate a density distribution over the images, while point-based models such as P2PNet [4] directly predict head locations and support joint counting and localization. More recently, MLP-based learning directions [5] have suggested that a simpler architecture, when combined with structured and interpretable geometric features, can still achieve effective learning performance.

B. The Problem Statement and Motivation

Crowd counting models trained on large-scale datasets often suffer performance degradation when applied to task-specific environments due to domain shifts, including variations in lighting conditions, camera perspectives, and scene characteristics. This domain gap significantly hinders model generalization to unseen scenarios [6], [7].
par In the context of the MTU broomball dataset, such domain discrepancies are pronounced. Unlike typical crowd-counting benchmark datasets that consist of outdoor surveillance or dense urban scenes, broomball games have different visibility due to weather and time of the day, dynamic player motion, overhead camera angles, and relatively sparse but structured crowds. As a result, pretrained models may fail to accurately estimate crowd density without task-specific adaptation or fine-tuning.

Another goal in this study was to benchmark transfer learning to simplified geometry-aware modeling for deployable, real-world crowd analytics systems. We take this comparison as important as with this, we can feed our model non-identifying information in the form of data storing masks and regions. This could allow published training data not to include any personal information or identifiable information. This could also potentially be used to reduce implicit biases in models that may be learned from texture and image-based regions. while also keeping the dataset.

C. Project statement

This research investigates crowd counting in images extracted from surveillance recordings of the Black Rink broomball field. The dataset exhibits both sparse and dense regions, where individuals appear at varying scales and are frequently occluded. Under these conditions, conventional detection-based approaches become unreliable due to occlusions and scale variations.

To address these challenges, we evaluate the performance of several deep learning models specifically designed for crowd counting, including P2PNet, CSRNet, and a feature-based preprocessed MLP model. Those models rely on density estimation and point-based regression rather than traditional image classification frameworks.

Furthermore, we leverage transfer learning techniques by fine-tuning pretrained models on annotated images from the surveillance recordings to improve counting accuracy and generalization performance in this domain-specific setting.

III. Data Sourcing and Processing

For our data, we will use recordings of previous broomball games at Michigan Technological University during Winter Carnival weekend. We only focused on the camera overlooking the Black Rink. We partitioned this data into a 60-20-20 train-test-validation split. The data for our transfer learning model comprises two games with 100 frames per game. Each frame was taken at a 30-second interval. For the MLP, we chose to only use the first game, as it was cloudy and provided no shadows, allowing a simpler image preprocessing.

A. Annotated Data Structure

For our input data, we included an inclusion zone mask that masked the area in which people would be both watching and playing the game. This was chosen to exclude passersby from our participation-based headcount. We then replaced the empty values resulting from that mask with the mean color of the previous image. This was chosen to reduce the drop-off in the average color of the image overall, which will be used in the preprocessed MLP.

We will be using supervised learning throughout this process, marking heads as the point of reference as per the common architecture. For this, if someone's head is not entirely in the image, they will not be counted in the overall count.

Fig. 1. Annotated Frame Example

B. MLP Preprocessing

To preprocess the MLP, we used a background subtraction. In this way, to account for some variations in time, we converted the BGR input to HSV to homogenize the value of the background to that of our sample image. This helps the non-uniformity of brightness across our different time sample images. Afterwards, we subtracted the sample image from the homogenized background in HSV to get a map of differences between the two images.

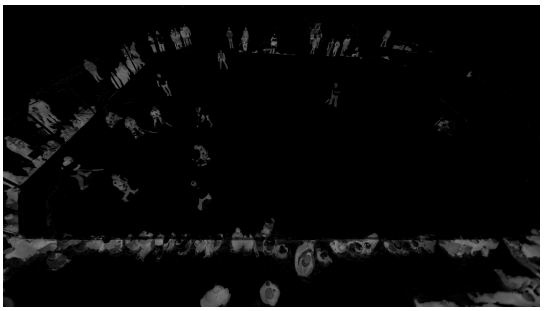

Fig. 2. Background Subtraction Example, Grayscale

We then furthered this higher signal-to-noise ratio image with OpenCV. Mainly, then using a global threshold for 20-25 pixels, followed by a dilation mask of the filled contours with a 5 by 5 kernel. The dilation of the masked contour area is done to fill any gaps made by the initial contours, which helps group close people together. We then compare this mask to our annotated data, labeling the specific contours by the number of heads that they contained. This in itself allowed our 100 frames to become 6557 pixel grouping data, which will be highlighted more in the next section. Below highlights where Figure 2 sees a grouping of people.

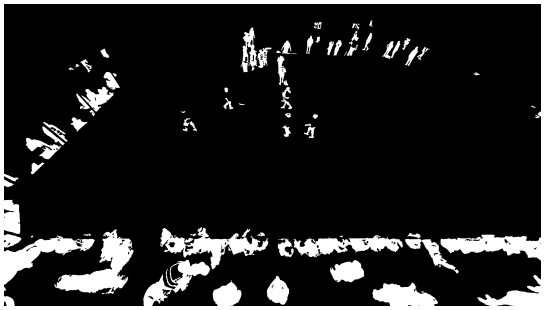

Fig. 3. Highlight of all output contours

IV. Implementation and Setup

A. P2PNet

P2PNet is a purely point-based crowd counting model that directly predicts head points, allowing counting and localization to be performed jointly. The model contains three main parts: a VGG16 backbone for feature extraction, an FPN for multi-scale feature fusion, and two prediction heads for confidence classification and point regression [4]. The FPN is important in our surveillance scenario because people appear at different scales due to camera angle and distance, while the point-based formulation makes the model especially suitable for our dataset, since our annotations are stored directly as head points.

During training, P2PNet uses one-to-one Hungarian matching to associate predicted points with ground-truth head points. This matching strategy is a key part of the model, because it enables direct point-level supervision. As a result, P2PNet is more naturally aligned with our annotation format and provides an interpretable way to perform both counting and approximate localization.

In our implementation, the input image is first processed by the VGG-16 backbone, then the extracted features are fused by the FPN, and finally, the prediction heads generate point candidates for counting and localization. To adapt the model to our small broomball dataset, we froze the backbone and fine-tuned the FPN and prediction heads. This helps retain general image features while allowing the specific layers to learn scene-dependent characteristics such as snow background and lighting variations.

The default pretrained model produced a MAE of 33.85, while the fine-tuned model reduced the MAE to 6.20. This shows that the default model was not matched to our target scene, likely due to domain differences between the original training data and our surveillance images. After fine-tuning, the model adapted well to the scene characteristics of our dataset, and the large reduction in MAE indicates that transfer learning was highly effective for this task.

B. CSRNet

1) Model Description: CSRNet is a suitable model for this task as it can effectively handle scale variation and dense crowd regions. It is a deep CNN that uses the first ten layers of VGG-16 as the front-end feature extractor and six dilated convolutional layers as the back-end for density map estimation [3]. The dilated convolutions avoid introducing additional pooling operations while expanding their receptive field, preserving higher spatial resolution in the feature maps compared to conventional architectures. This allows the model to capture the broader contextual information, making it reliable in distinguishing closely packed individuals in dense scenes. To establish a baseline, we first evaluated CSRNet using the pre-trained weights. The model was able to generate a density map that captures the general spatial distribution and approximate locations of the crowd in the scene. However, the estimated count was differ than the actual number of individuals. The discrepancy is likely due to domain differences between the pre-training dataset (ShanghaiTech [8]) and our surveillance data, such as variations in lighting conditions, camera perspective, resolution, and seasonal environment.

To mitigate this issue, we fine-tuned the model using annotated frames from the surveillance recordings. Given the limited size of the dataset, we froze the front-end VGG-16 layers during training and updated only the back-end convolutional layers and output layer. This approach preserves the general feature extraction capability learned from large-scale datasets while allowing the model to adapt to domain-specific characteristics. Freezing the early layers also reduces the number of trainable parameters, which reduces the computational cost.

2) Training Setup: For CSRNet training, we incorporated an additional broomball game recording captured during Winter Carnival, resulting in a total dataset of 200 frames. After annotation, the point-based head labels were converted into continuous density maps using a Gaussian kernel, as shown in Fig. 4. This process produces a heatmap representation in which each annotated point contributes a localized distribution whose spread is controlled by the Gaussian parameter (γ). The choice of this parameter directly affects the resulting density map: a larger γ produces smoother distributions that emphasize global structure, while a smaller γ yields sharper peaks that better preserve precise spatial localization. In our implementation, we use a sigma value of 12 to match the lower-resolution output of CSRNet. This transformation ensures consistency between the model output and the ground truth during training. Due to the small dataset size, careful selection of training hyperparameters are essential. We used the learning rate of $1*10^-7$ to prevent large updates that could destabilize the pretrained weights and lead to overfitting. The original model use 500 epochs to train the model; for this implementation, we use only 50 epochs. An early stopping strategy is employed based on validation mean absolute error (MAE). Early stopping halts training once performance on the validation set ceases to improve in 5 consecutive epochs, preventing the model from memorizing the training data and degrading generalization performance.

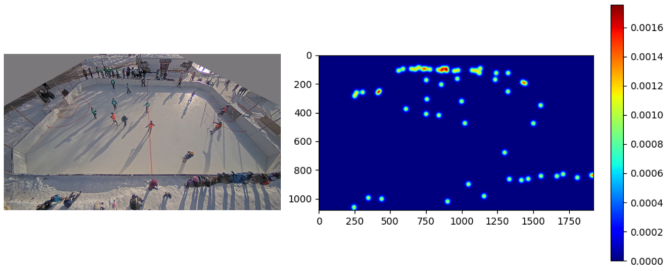

Fig. 4. Example annotation to heatmap conversion for CSRNet training input.

C. Preprocessed MLP

Our final model is that of an MLP, with many inputs derived from physical quantities found through a commonplace contour detection. For the construction of the MLP, it will have three hidden layers, each with a width of 256. Followed by a batch normalization layer, a dropout layer of thirty-five percent, and ReLU activation for each layer for its nonlinearity. Due to the nature of the system, we will also calculate loss with L2 regularization with a weight of 0.01. The initial weights of the system is further defined by He Normalization Initialization. [9].

We will use OpenCV to extract many features from the image, which we can compare against the model. The data that will be fed from these segments to the base inputs for the perceptron includes;

- Centroid: The center location of the contour. Given as X and Y pixel locations
- Object Area: The number of pixels taken by a contour.
- Solidity: Ratio percentage of contour area compared to the convex hull [10].

We will then be exploring an exploitation of geometry to account for wide-angle distortion. As a pretext to test if including it will improve the base perceptron [5]. For this, we will include the following to exploit the system's data, which only relies on the assumption that the camera will not move. This will make the inputs of our tuned exploitation a derivation from the previous results.

- Centroid displacement: The center location of the contour. Now given as a radius and an angle from the pixel approximated directly below the camera. Induces symmetry based on how effective pixel area decreases as we get further from the camera.
- Normalized Dot Product: The dot product of the best-fit line [10] of the contour with respect to the line perpendicular to the centroid displacement vector. This will measure how much of our pixel area is able to be taken at a similar real-world size.
- Object Area and Solidity: Same as before.

V. Results and Comparison

In table I, we evaluated the performance across three models, where each model has different trade-offs between accuracy, computational cost, and generalization in crowd

counting tasks. CSRNet achieves the best overall performance, with the lowest base and second-lowest fine-tuned MAE, due to its deep convolutional architecture that predicts high-quality density maps. This makes it particularly effective in sparse or moderately crowded outdoor scenes. However, it is computationally intensive, requires careful parameter tuning (e.g., dilation rate and learning rate), and depends on Gaussian-based preprocessing, which may introduce a mismatch with point annotations. P2PNet, in contrast, directly predicts head locations, aligning more naturally with point annotations and supporting both counting and localization, thus seeing a significant improvement in accuracy after fine-tuning and has the lowest fine-tuning model MAE. While it offers strong accuracy and good generalization, its performance still relies on labeled data and fine-tuning and has a relatively high computational cost due to excessive parameters. MLP offers the fastest computation and the lowest structural complexity, making it efficient and easy to deploy. The geometric data also effectively increases training samples. However, its lower accuracy and weaker generalization largely stem from its reliance on manual data preprocessing and feature extraction. This process often reduces or omits important information—such as color, spatial relationships, and environmental context—leading to a loss of rich visual features that are critical for accurate crowd counting. Overall, CSRNet excels in accuracy and robustness, P2PNet balances localization and performance, and MLP prioritizes efficiency at the cost of accuracy.

TABLE I
THE EVALUATION MATRICES TABLE FOR CSRNET, P2PNET AND MLP MODELS

Evaluation Matrices	**Models**		
	CSRNet	***P2PNet***	***MLP***
Basic Model MAE	12.50	33.85	38.01
Fine-tuned/Advanced Model MAE	7.40	6.20	24.65
Generalization	Medium	High	Very Low
Computational Cost	Medium	High	Very Low
Traceability	Low	Low	High
Structure Complexity	High	High	low
Overall Performance	High	High	Medium

VI. CONCLUSION

In investigating the performance and feasibility of applying transfer learning to a CNN-based crowd-counting model for a domain-specific task, namely, estimating attendance at broomball games. In parallel, we developed a geometry-encoded MLP trained on manually extracted, preprocessed features as a lightweight alternative. The results demonstrate that each approach presents distinct trade-offs in terms of accuracy, computational cost, and generalization capability.

Overall, CSRNet shows stronger performance for outdoor scenes with relatively sparse and structured crowds, making it a suitable baseline for practical crowd-counting applications in such environments. For model adaptation, our findings suggest that transfer learning with a limited amount of locally annotated data is effective, particularly when fine-tuning is restricted to task-specific layers. This strategy helps mitigate overfitting while maintaining or improving prediction accuracy.

VII. FUTURE WORK

Highlighting future work, there could be a push to increase the symmetry even further, especially in more controlled environments than our outdoor testing. An example from here could be theta measurements collapsing into solely the first quadrant.

Another improvement could be the initial filter; as for this data set, the orange pull-over jerseys given to players are removed as a byproduct of our filter. It could also be good to highlight how putting this filtered data through the other benchmarks would affect their performance, as it effectively increases the signal-to-noise ratio. It could also be good to see how a dataset that is more body-labeled or any body part would affect the performance of a body counting model, as it is much easier to find a person's torso with image processing than a head, especially when pixel area is in play.

The last potential experiment is that our MLP is simplistic in nature, only testing the increase in quality of input, but refining the model itself could provide compounding returns.

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SELF-DECLARATION

Self-Declaration: We, the undersigned, confirm that this proposal is a group project. Division of tasks:

• Nolan Datema: Constructing and training the geometry encoded MLP framework, image processing, image annotation, and report writing

• Marcus Wu: Training and fine-tuning the CSRNet, data annotation, and report writing.

• Xinying Li: Training and fine-tuning the P2PNet, data annotation and report writing.